

Determine smallest k making buy- k revenue finitely approximable by bundling

ChatGPT (gpt-5.6-sol)*

Solution generated on 2026-08-24

Abstract

This paper gives a complete solution to the finiteness-threshold problem for bundling approximations to additive buy- k mechanisms. For one item, bundling is exactly optimal for every k . For every $n \geq 2$, buy-one mechanisms have an unbounded revenue advantage over grand-bundle pricing, whereas every buy- k mechanism with $k \geq 2$ has revenue at most

$$16e 4^n n!$$

times the optimal grand-bundle revenue. When $k \geq n$, the upper bound improves to $8n^2$. Consequently,

$$k_{\min}(n) = \begin{cases} 1, & n = 1, \\ 2, & n \geq 2. \end{cases}$$

Thus no gap remains concerning the exact finiteness threshold. The numerical approximation constants are not claimed to be optimal.

1 The open problem

This paper addresses an open problem posed by Sepehr Assadi, Vikram Kher, George Li, Ariel Schwartzman in *Fine-Grained Buy-Many Mechanisms Are Not Much Better Than Bundling* [1] (Economics and Computation (EC), 2023), Section 5 (Conclusion), page 21; also implicit in Question 1 in Section 1 (Introduction), page 4, and reiterated after Lemma 8 in Section 3, page 10.. The website catalogue entry is problem 6 (W4383533365_p0) of the [Open Problems in Operations Research](#) collection.

Determine, as a function of n , the smallest integer k such that $\Gamma(n, k) < \infty$, i.e., such that there exists a finite function $f(n, k)$ satisfying

$$\text{Buy}_k \text{Rev}(D) \leq f(n, k) \cdot \text{BRev}(D) \quad \text{for all distributions } D \text{ over additive valuations on } n \text{ items.}$$

Equivalently, characterize the threshold on k (in terms of n) at which the worst-case ratio $\sup_D \text{Buy}_k \text{Rev}(D) / \text{BRev}(D)$ becomes finite, and more generally determine the tradeoff of this ratio as a function of (n, k) .

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.9 write-up; run `run_20260824_024352_e5b1e180, ec-2it-ultra`); two-iteration run, outcome from iteration 1. Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

k Incentive Compatibility

2 Model and main result

There are n heterogeneous items. A buyer has an additive, nonnegative valuation

$$x = (x_1, \dots, x_n) \in \mathbb{R}_+^n.$$

No independence assumption is imposed on the coordinates of x . A lottery is represented by its vector of marginal allocation probabilities

$$q = (q_1, \dots, q_n) \in [0, 1]^n,$$

and its expected value to type x is $x \cdot q$.

If two lotteries are purchased independently, their combined allocation is

$$q \oplus q' = \left(1 - (1 - q_\ell)(1 - q'_\ell)\right)_{\ell=1}^n.$$

More generally,

$$\left(\bigoplus_{h=1}^r q^h\right)_\ell = 1 - \prod_{h=1}^r (1 - q_\ell^h).$$

Repeated purchases of the same menu option are permitted.

A menu consists of options $(q, p) \in [0, 1]^n \times \mathbb{R}_+$ and contains the outside option $(0, 0)$. A mechanism is equipped with a measurable selection

$$x \mapsto (q(x), P(x))$$

of a utility-maximizing menu option. We assume throughout that the selected maximum exists; this is automatic for the finite menus used in the lower-bound construction.

For a positive integer k , the mechanism is *buy- k incentive compatible* if, for every type x , every $1 \leq r \leq k$, and every collection of menu options

$$(q^1, p^1), \dots, (q^r, p^r),$$

possibly with repetitions, one has

$$x \cdot q(x) - P(x) \geq x \cdot \left(\bigoplus_{h=1}^r q^h\right) - \sum_{h=1}^r p^h. \quad (1)$$

Taking the outside option as the deviation gives individual rationality:

$$0 \leq P(x) \leq x \cdot q(x) \leq \|x\|_1. \quad (2)$$

The same results hold under the weaker convention in which (1) is required only for types in the support of the distribution; indeed, the upper-bound proofs use incentive compatibility only at representative types drawn from sets of positive probability.

Let D be an arbitrary Borel probability distribution on $\mathbb{R}_+^n$, and let $X \sim D$. Write

$$S(X) = \|X\|_1.$$

The optimal grand-bundle revenue is defined as the supremum

$$\text{BRev}(D) = \sup_{z \geq 0} z \Pr[S(X) \geq z]. \quad (3)$$

The use of a supremum is necessary because an optimal posted price need not exist for an arbitrary distribution.

Let $\text{Buy}_k \text{Rev}(D)$ denote the supremum of $\mathbb{E}[P(X)]$ over all buy- k incentive-compatible mechanisms. Define

$$\Gamma(n, k) = \sup_{\substack{D \text{ on } \mathbb{R}_+^n \\ 0 < \text{BRev}(D) < \infty}} \frac{\text{Buy}_k \text{Rev}(D)}{\text{BRev}(D)}. \quad (4)$$

For every bundle price z , the menu consisting of $(0, 0)$ and $(\mathbf{1}, z)$ is buy- k incentive compatible for every k : purchasing the bundle more than once does not increase the allocation and only increases the payment. Taking the supremum over z therefore gives

$$\text{Buy}_k \text{Rev}(D) \geq \text{BRev}(D). \quad (5)$$

Theorem 1 (Complete classification). *For every positive integer k ,*

$$\Gamma(1, k) = 1. \quad (6)$$

For every $n \geq 2$,

$$\Gamma(n, 1) = \infty, \quad (7)$$

whereas, for every $k \geq 2$,

$$1 \leq \Gamma(n, k) \leq 16e^4 n!. \quad (8)$$

For every $n \geq 1$ and $k \geq n$,

$$\Gamma(n, k) \leq 8n^2. \quad (9)$$

Consequently, if

$$k_{\min}(n) = \min\{k \geq 1 : \Gamma(n, k) < \infty\},$$

then

$$k_{\min}(n) = \begin{cases} 1, & n = 1, \\ 2, & n \geq 2. \end{cases} \quad (10)$$

The ratio in (4) is restricted to $0 < \text{BRev}(D) < \infty$. This causes no loss in the classification. If $\text{BRev}(D) = 0$, then $S(X) = 0$ almost surely: otherwise some event $\{S(X) \geq 1/m\}$ would have positive probability and would give positive bundle revenue. Equation (2) then forces every individually rational mechanism to have zero revenue. If $\text{BRev}(D) = \infty$, the asserted upper bounds are valid in the extended-real sense but do not define a literal ratio of two infinite quantities.

3 A packing lemma for buy-two deviations

For $u \in [0, 1]$, define

$$\phi(u) = 2u - u^2. \quad (11)$$

Thus $q \oplus q = \phi(q)$, with ϕ applied coordinatewise.

Lemma 2 (Buy-two packing lemma). *Let*

$$q_0 = 0, \quad q_1, \dots, q_N \in [0, 1]^n$$

be a finite sequence, and let $x_i \in \mathbb{R}_+^n \setminus \{0\}$ for $1 \leq i \leq N$. Put

$$w_i = \frac{x_i}{\|x_i\|_1}$$

and

$$d_i = \min_{0 \leq r, s < i} w_i \cdot (q_i - (q_r \oplus q_s)), \quad (12)$$

where $r = s$ is allowed. Then

$$\sum_{i=1}^N (d_i)_+ < e 4^n n!. \quad (13)$$

The same bound holds for a countable sequence.

Proof. Fix $\varepsilon \in (0, 1]$. Define breakpoints recursively by

$$a_0 = 0, \quad a_{h+1} = \min\{1, \phi(a_h) + \varepsilon\}, \quad (14)$$

and stop at the first index H for which $a_H = 1$. These breakpoints form H half-open intervals partitioning $[0, 1]$, with the final interval closed at 1.

The function ϕ is nondecreasing on $[0, 1]$. If u and v lie in an interval beginning at a_h , then

$$u \leq a_{h+1} \leq \phi(a_h) + \varepsilon \leq \phi(v) + \varepsilon. \quad (15)$$

We next prove the explicit bound

$$H \leq 4(1 + \log(1/\varepsilon)). \quad (16)$$

If $a_h \leq 1/2$ and the step from a_h is nonterminal, then

$$a_{h+1} = 2a_h - a_h^2 + \varepsilon \geq \frac{3}{2}a_h + \varepsilon. \quad (17)$$

After r consecutive nonterminal steps that remain below $1/2$, induction gives

$$a_r \geq 2\varepsilon \left[\left(\frac{3}{2} \right)^r - 1 \right]. \quad (18)$$

It follows that the number of steps before reaching $1/2$, or terminating earlier, is at most

$$\frac{\log(1 + 1/(4\varepsilon))}{\log(3/2)} + 1. \quad (19)$$

Once $a_h \geq 1/2$, put $b_h = 1 - a_h$, so that $0 \leq b_h \leq 1/2$. On every nonterminal step,

$$b_{h+1} = b_h^2 - \varepsilon \leq b_h^2 \leq \frac{1}{2}b_h. \quad (20)$$

Once $b_h^2 \leq \varepsilon$, the next step is terminal. Starting from $b_h \leq 1/2$, the number of additional steps is therefore at most

$$\frac{\log(1/\varepsilon)}{2 \log 2} + 2. \quad (21)$$

Combining (19) and (21), and writing $t = \log(1/\varepsilon)$, gives

$$H \leq 3 + \frac{\log(1 + e^t/4)}{\log(3/2)} + \frac{t}{2 \log 2}. \quad (22)$$

Since

$$1 + \frac{e^t}{4} \leq \frac{5}{4}e^t,$$

we obtain

$$H \leq 3 + \frac{\log(5/4)}{\log(3/2)} + \left(\frac{1}{\log(3/2)} + \frac{1}{2 \log 2} \right) t. \quad (23)$$

Here $\log(5/4) < \log(3/2)$. Moreover,

$$\log(3/2) = \int_1^{3/2} \frac{du}{u} \geq \frac{1}{3}, \quad \log 2 = \int_1^2 \frac{du}{u} \geq \frac{1}{2}.$$

Thus the constant term in (23) is at most 4, and the coefficient of t is at most 4. This proves (16).

Partition $[0, 1]^n$ by assigning an interval from the preceding partition to each coordinate. There are at most

$$\left[4(1 + \log(1/\varepsilon)) \right]^n \quad (24)$$

coordinatewise signatures.

Suppose that $i < j$ and that q_i, q_j have the same signature. Applying (15) in every coordinate gives

$$q_{j,\ell} \leq \phi(q_{i,\ell}) + \varepsilon, \quad \ell = 1, \dots, n. \quad (25)$$

Since the repeated pair (q_i, q_i) is admissible in (12),

$$\begin{aligned} d_j &\leq w_j \cdot (q_j - (q_i \oplus q_i)) \\ &= w_j \cdot (q_j - \phi(q_i)) \leq \varepsilon \sum_{\ell=1}^n w_{j,\ell} = \varepsilon. \end{aligned} \quad (26)$$

Consequently, each signature contains at most one index satisfying $d_i > \varepsilon$, and hence

$$\#\{i : d_i > \varepsilon\} \leq \left[4(1 + \log(1/\varepsilon)) \right]^n. \quad (27)$$

Using the pair $(q_0, q_0) = (0, 0)$ in (12), we also have

$$d_i \leq w_i \cdot q_i \leq 1. \quad (28)$$

For nonnegative numbers z_i , the layer-cake identity states that

$$\sum_i z_i = \int_0^\infty \#\{i : z_i > t\} dt. \quad (29)$$

For a finite family this follows by integrating each indicator $\mathbf{1}_{\{t < z_i\}}$; for a countable family it follows from Tonelli's theorem applied to the nonnegative measurable function on the product of counting measure and Lebesgue measure. Applying (29) to $z_i = (d_i)_+$, and using (27)–(28), yields

$$\begin{aligned} \sum_{i=1}^N (d_i)_+ &= \int_0^1 \#\{i : d_i > t\} dt \\ &\leq 4^n \int_0^1 (1 + \log(1/t))^n dt. \end{aligned} \quad (30)$$

With the substitution $t = e^{-s}$,

$$\begin{aligned} \int_0^1 (1 + \log(1/t))^n dt &= \int_0^\infty (1 + s)^n e^{-s} ds \\ &= \sum_{r=0}^n \binom{n}{r} \int_0^\infty s^r e^{-s} ds \\ &= \sum_{r=0}^n \binom{n}{r} r! \\ &= n! \sum_{j=0}^n \frac{1}{j!} < e n!. \end{aligned} \quad (31)$$

This proves (13). For a countable sequence, apply (13) to every finite prefix and then take the increasing limit of the partial sums of the nonnegative quantities $(d_i)_+$. $\square$

4 Revenue conversion and the buy-two upper bound

Proposition 3. *For every distribution D on $\mathbb{R}_+^n$ and every buy-two incentive-compatible mechanism M ,*

$$\text{Rev}(M, D) \leq 16e 4^n n! \text{BRev}(D). \quad (32)$$

Proof. The assertion is immediate in the extended-real sense if $\text{BRev}(D) = \infty$, and the case $\text{BRev}(D) = 0$ was addressed after Theorem 1. Assume

$$0 < B \doteq \text{BRev}(D) < \infty.$$

Let $P(x)$ be the selected payment and $S(x) = \|x\|_1$. Fix

$$0 < a < b < \infty.$$

For $t \in \mathbb{Z}$, define the truncated geometric payment band

$$E_t = \{x : a \leq P(x) < b\} \cap \{x : a \cdot 4^t \leq P(x) < a \cdot 4^{t+1}\}. \quad (33)$$

Only finitely many of these sets are nonempty. Put

$$R_t = \mathbb{E}[P(X)\mathbf{1}_{\{X \in E_t\}}]. \quad (34)$$

One of the two parity classes of t carries at least half of the total truncated revenue. Retain that parity and discard all null bands.

Let the retained band indices be

$$t_1 < t_2 < \cdots < t_N.$$

For every i , the random variable S is finite on E_{t_i} , and by (2),

$$S(x) \geq P(x) \geq a \cdot 4^{t_i} > 0 \quad (x \in E_{t_i}).$$

Thus

$$0 < \sigma_i = \operatorname{ess\,inf}_{x \in E_{t_i}} S(x) < \infty, \quad (35)$$

where the essential infimum is relative to D restricted to E_{t_i} . For an arbitrary $\eta > 0$, choose a representative

$$x_i \in E_{t_i}$$

such that

$$S_i = \|x_i\|_1 \leq (1 + \eta)\sigma_i. \quad (36)$$

Such a representative exists by the definition of the essential infimum. Write its selected menu option as

$$(q_i, p_i) = (q(x_i), P(x_i)).$$

Also put $q_0 = 0$ and $p_0 = 0$.

Because the retained indices have the same parity, $t_i - t_r \geq 2$ whenever $r < i$. Hence

$$p_r < a \cdot 4^{t_r+1} \leq a \cdot 4^{t_i-1} \leq \frac{p_i}{4}. \quad (37)$$

The same inequality is valid for $r = 0$. Therefore, for every $0 \leq r, s < i$,

$$p_r + p_s < \frac{p_i}{2}. \quad (38)$$

Buy-two incentive compatibility at x_i gives

$$x_i \cdot q_i - p_i \geq x_i \cdot (q_r \oplus q_s) - p_r - p_s.$$

Consequently,

$$x_i \cdot (q_i - (q_r \oplus q_s)) \geq p_i - p_r - p_s > \frac{p_i}{2}. \quad (39)$$

With $w_i = x_i/S_i$, define

$$d_i = \min_{0 \leq r, s < i} w_i \cdot (q_i - (q_r \oplus q_s)). \quad (40)$$

Equation (39) implies

$$d_i > \frac{p_i}{2S_i} > 0. \quad (41)$$

By (36), almost every $x \in E_{t_i}$ satisfies

$$S(x) \geq \sigma_i \geq \frac{S_i}{1 + \eta}. \quad (42)$$

Posting the grand-bundle price $S_i/(1 + \eta)$ therefore sells to every type in E_{t_i} , apart from a null set. By the definition of B ,

$$B \geq \frac{S_i}{1 + \eta} \Pr \left[S(X) \geq \frac{S_i}{1 + \eta} \right] \geq \frac{S_i}{1 + \eta} \Pr[E_{t_i}],$$

and hence

$$\Pr[E_{t_i}] \leq \frac{(1 + \eta)B}{S_i}. \quad (43)$$

For every $x \in E_{t_i}$,

$$P(x) < a \cdot 4^{t_i+1} \leq 4p_i. \quad (44)$$

Combining (41), (43), and (44), we obtain

$$\begin{aligned} R_{t_i} &\leq 4p_i \Pr[E_{t_i}] \\ &\leq 4(1 + \eta)B \frac{p_i}{S_i} \\ &< 8(1 + \eta)B d_i. \end{aligned} \quad (45)$$

The packing lemma now gives

$$\sum_{i=1}^N R_{t_i} < 8(1 + \eta)e 4^n n! B. \quad (46)$$

The retained parity carries at least half of the truncated revenue, so

$$\mathbb{E} [P(X) \mathbf{1}_{\{a \leq P(X) < b\}}] \leq 16(1 + \eta)e 4^n n! B. \quad (47)$$

Since (47) holds for every $\eta > 0$, letting $\eta \downarrow 0$ gives

$$\mathbb{E} [P(X) \mathbf{1}_{\{a \leq P(X) < b\}}] \leq 16e 4^n n! B. \quad (48)$$

For integers $m \geq 2$, set

$$Y_m = P(X) \mathbf{1}_{\{1/m \leq P(X) < m\}}.$$

The variables Y_m are nonnegative and measurable, satisfy $Y_m \uparrow P(X)$ pointwise, and are covered by (48) with $a = 1/m$ and $b = m$. The monotone convergence theorem, in the form $\mathbb{E}[Y_m] \uparrow \mathbb{E}[P(X)]$ for a pointwise increasing sequence of nonnegative measurable random variables, therefore yields

$$\text{Rev}(M, D) = \mathbb{E}[P(X)] \leq 16e 4^n n! B.$$

This proves (32), including the possibility that the revenue was initially allowed to be infinite. $\square$

Every buy- k mechanism with $k \geq 2$ is buy-two incentive compatible. Taking the supremum over such mechanisms in (32) proves that

$$\text{Buy}_k \text{Rev}(D) \leq 16e 4^n n! \text{BRev}(D), \quad k \geq 2. \quad (49)$$

5 The sharper bound when $k \geq n$

We first establish a telescoping packing estimate for deviations using at most n earlier options.

Lemma 4 (Buy- n telescoping bound). *Let*

$$q_0 = 0, \quad q_1, \dots, q_N \in [0, 1]^n,$$

and let $w_i \in \mathbb{R}_+^n$ satisfy $\sum_{\ell=1}^n w_{i,\ell} = 1$. Define

$$d_i^{(n)} = \min_{0 \leq r_1, \dots, r_n < i} w_i \cdot \left(q_i - \bigoplus_{h=1}^n q_{r_h} \right), \quad (50)$$

where indices may repeat. Then

$$\sum_{i=1}^N (d_i^{(n)})_+ \leq n. \quad (51)$$

Proof. For $i \geq 1$ and $1 \leq \ell \leq n$, put

$$m_{i,\ell} = \max_{0 \leq r < i} q_{r,\ell}. \quad (52)$$

The maximum exists because there are finitely many predecessors. For every coordinate ℓ , choose an index $r(i, \ell) < i$ attaining this maximum; the selected indices may repeat. Let

$$Q_i = \bigoplus_{\ell=1}^n q_{r(i,\ell)}.$$

For every coordinate j , one of the combined lotteries is $q_{r(i,j)}$, so

$$Q_{i,j} \geq q_{r(i,j),j} = m_{i,j}. \quad (53)$$

It follows that

$$\begin{aligned} d_i^{(n)} &\leq w_i \cdot (q_i - Q_i) \\ &\leq \sum_{\ell=1}^n w_{i,\ell} (q_{i,\ell} - m_{i,\ell}) \\ &\leq \sum_{\ell=1}^n (q_{i,\ell} - m_{i,\ell})_+. \end{aligned} \quad (54)$$

The final inequality holds termwise: if $q_{i,\ell} - m_{i,\ell} < 0$, its weighted contribution is nonpositive, and otherwise $w_{i,\ell} \leq 1$.

The running maxima obey

$$m_{i+1,\ell} = \max\{m_{i,\ell}, q_{i,\ell}\},$$

and hence

$$(q_{i,\ell} - m_{i,\ell})_+ = m_{i+1,\ell} - m_{i,\ell}. \quad (55)$$

Since the right-hand side of (54) is nonnegative,

$$\begin{aligned} \sum_{i=1}^N (d_i^{(n)})_+ &\leq \sum_{\ell=1}^n \sum_{i=1}^N (m_{i+1,\ell} - m_{i,\ell}) \\ &= \sum_{\ell=1}^n (m_{N+1,\ell} - m_{1,\ell}) \\ &\leq n, \end{aligned}$$

because $m_{1,\ell} = q_{0,\ell} = 0$ and $m_{N+1,\ell} \leq 1$. $\square$

Proposition 5. *For every $k \geq n$,*

$$\text{Buy}_k \text{Rev}(D) \leq 8n^2 \text{BRev}(D). \quad (56)$$

Proof. It suffices to consider a buy- n incentive-compatible mechanism. Assume $0 < B \doteq \text{BRev}(D) < \infty$, since the degenerate cases have already been handled.

Fix $0 < a < b < \infty$ and set $\rho = 2n$. Partition the truncated positive payments into bands

$$E_t = \{x : a \leq P(x) < b\} \cap \{x : a\rho^t \leq P(x) < a\rho^{t+1}\}. \quad (57)$$

Retain a parity of t carrying at least half of the truncated revenue. As in the proof of Proposition 3, choose from every retained band of positive probability a representative x_i satisfying

$$S_i \doteq \|x_i\|_1 \leq (1 + \eta) \operatorname{ess\,inf}_{x \in E_i} S(x), \quad (58)$$

where E_i denotes the band containing x_i , and write its selected option as (q_i, p_i) . Order the representatives by increasing band index, and include $(q_0, p_0) = (0, 0)$.

If $r < i$, the parity separation gives

$$p_r < \frac{p_i}{\rho} = \frac{p_i}{2n}. \quad (59)$$

Thus any n earlier options, including repetitions and outside-option padding, have total price strictly less than $p_i/2$. Buy- n incentive compatibility gives, for every $0 \leq r_1, \dots, r_n < i$,

$$x_i \cdot \left(q_i - \bigoplus_{h=1}^n q_{r_h} \right) > \frac{p_i}{2}. \quad (60)$$

With $w_i = x_i/S_i$, the gap in (50) therefore satisfies

$$d_i^{(n)} > \frac{p_i}{2S_i} > 0. \quad (61)$$

The bundle-price argument based on (58) gives

$$\Pr[E_i] \leq \frac{(1 + \eta)B}{S_i}. \quad (62)$$

Every payment in E_i is less than $\rho p_i = 2np_i$. Hence its band revenue R_i satisfies

$$\begin{aligned} R_i &\leq 2np_i \Pr[E_i] \\ &\leq 2n(1+\eta)B \frac{p_i}{S_i} \\ &< 4n(1+\eta)B d_i^{(n)}. \end{aligned} \tag{63}$$

By Lemma 4,

$$\sum_{\text{retained } i} R_i \leq 4n^2(1+\eta)B. \tag{64}$$

Restoring the discarded parity, letting $\eta \downarrow 0$, and then applying monotone convergence to the nested truncations $\{1/m \leq P < m\}$ gives

$$\text{Rev}(M, D) \leq 8n^2B.$$

Taking the supremum over buy- k mechanisms proves (56), because every buy- k mechanism with $k \geq n$ is buy- n incentive compatible. $\square$

6 An unbounded buy-one gap for every $n \geq 2$

It suffices first to construct a two-item example; the extension to $n > 2$ appears at the end of the section.

Fix an integer $L \geq 3$, and let

$$m = (L+1)^2 - 1.$$

Enumerate the $(L+1)^2$ grid points in $[0, 1]^2$ by

$$g_j = \left(\frac{a_j}{L}, \frac{b_j}{L} \right), \quad j = a_j(L+1) + b_j, \tag{65}$$

where $0 \leq a_j, b_j \leq L$. Thus $j = 0, \dots, m$, and $g_0 = (0, 0)$.

Consider the finite menu consisting of the outside option $(g_0, 0)$ and, for $1 \leq j \leq m$,

$$(q_j, p_j) = \left(g_j, \frac{1}{2}L^{j-1} \right). \tag{66}$$

Define

$$y_j = (L+1-b_j, 1), \quad x_j = L^j y_j, \tag{67}$$

and let

$$\xi_j = \|x_j\|_1 = L^j(L+2-b_j). \tag{68}$$

Lemma 6. *For every $1 \leq j \leq m$, option j strictly dominates every earlier option for type x_j .*

Proof. Fix $0 \leq \ell < j$. The enumeration in (65) implies either $a_\ell = a_j$ and $b_\ell < b_j$, or $a_\ell < a_j$.

If $a_\ell = a_j$, then

$$L(g_j - g_\ell) \cdot y_j = b_j - b_\ell \geq 1.$$

If $a_\ell < a_j$, then

$$\begin{aligned} L(g_j - g_\ell) \cdot y_j &= (a_j - a_\ell)(L+1-b_j) + (b_j - b_\ell) \\ &\geq (L+1-b_j) + (b_j - L) = 1. \end{aligned}$$

Therefore, in all cases,

$$(g_j - g_\ell) \cdot y_j \geq \frac{1}{L}, \quad (69)$$

and hence

$$x_j \cdot (g_j - g_\ell) = L^j y_j \cdot (g_j - g_\ell) \geq L^{j-1}. \quad (70)$$

For every $\ell < j$, including $\ell = 0$,

$$p_j - p_\ell \leq p_j = \frac{1}{2} L^{j-1} < L^{j-1}. \quad (71)$$

For the outside option, the first inequality in (71) holds with equality, but the final strict inequality remains valid. Combining (70) and (71),

$$(x_j \cdot g_j - p_j) - (x_j \cdot g_\ell - p_\ell) > 0.$$

Thus option j strictly dominates every earlier option. $\square$

Because the menu is finite, every type has a utility-maximizing option. Lemma 6 shows that every maximizer for type x_j has index at least j . Since the payments p_j increase with j , every such maximizer charges at least p_j . No favorable tie-breaking is required.

The bundle values ξ_j are strictly increasing. Indeed, if $b_j < L$, then $b_{j+1} = b_j + 1$, and

$$\frac{\xi_{j+1}}{\xi_j} = \frac{L(L+1-b_j)}{L+2-b_j} > 1, \quad (72)$$

because $L \geq 3$ and $L+1-b_j \geq 2$. If $b_j = L$, then the next grid row starts with $b_{j+1} = 0$, and

$$\frac{\xi_{j+1}}{\xi_j} = \frac{L(L+2)}{2} > 1. \quad (73)$$

Set $\xi_{m+1} = \infty$, with $1/\xi_{m+1} = 0$, and define a finite-support distribution D_L by

$$\Pr[X = x_j] = \frac{\xi_1}{\xi_j} - \frac{\xi_1}{\xi_{j+1}}, \quad 1 \leq j \leq m. \quad (74)$$

Strict monotonicity of the sequence (ξ_j) makes every probability positive, and telescoping gives total mass one. Moreover,

$$\Pr[\|X\|_1 \geq \xi_j] = \sum_{h=j}^m \left(\frac{\xi_1}{\xi_h} - \frac{\xi_1}{\xi_{h+1}} \right) = \frac{\xi_1}{\xi_j}. \quad (75)$$

We claim that

$$\text{BRev}(D_L) = \xi_1. \quad (76)$$

Every bundle price ξ_j earns exactly ξ_1 by (75). Conversely, if $0 \leq z \leq \xi_1$, then its revenue is at most $z \leq \xi_1$. If

$$\xi_j < z \leq \xi_{j+1}$$

for some $j < m$, then

$$z \Pr[\|X\|_1 \geq z] = z \frac{\xi_1}{\xi_{j+1}} \leq \xi_1.$$

Finally, a price above ξ_m earns zero. This proves (76).

The menu in (66) is buy-one incentive compatible when each type is assigned any utility-maximizing option. Its revenue under D_L is at least

$$\sum_{j=1}^m p_j \Pr[X = x_j].$$

Using (74) and summation by parts, with $p_0 = 0$, gives

$$\begin{aligned} \frac{\text{Rev}(M, D_L)}{\xi_1} &\geq \sum_{j=1}^m p_j \left(\frac{1}{\xi_j} - \frac{1}{\xi_{j+1}} \right) \\ &= \sum_{j=1}^m \frac{p_j - p_{j-1}}{\xi_j}. \end{aligned} \tag{77}$$

All summands are nonnegative. Restrict the sum to grid points satisfying

$$1 \leq a_j \leq L, \quad 1 \leq b_j \leq L.$$

For these indices $j \geq 2$, and

$$p_j - p_{j-1} = \frac{1}{2} L^{j-2} (L - 1).$$

Using (68),

$$\frac{p_j - p_{j-1}}{\xi_j} = \frac{L - 1}{2L^2(L + 2 - b_j)}. \tag{78}$$

For every fixed $b_j \in \{1, \dots, L\}$, there are L permitted values of a_j . Therefore,

$$\begin{aligned} \frac{\text{Rev}(M, D_L)}{\text{BRev}(D_L)} &\geq \frac{L - 1}{2L} \sum_{b=1}^L \frac{1}{L + 2 - b} \\ &= \frac{L - 1}{2L} \sum_{r=2}^{L+1} \frac{1}{r}. \end{aligned} \tag{79}$$

The right-hand side diverges as $L \rightarrow \infty$. It follows that

$$\Gamma(2, 1) = \infty. \tag{80}$$

For $n > 2$, append $n - 2$ zero coordinates to every type and every allocation. This preserves all utilities, selected payments, and grand-bundle values. Hence

$$\Gamma(n, 1) = \infty \quad (n \geq 2). \tag{81}$$

7 The one-item case

Proposition 7. *For every distribution D on $\mathbb{R}_+$ and every menu with a measurable utility-maximizing selection,*

$$\text{Rev}(M, D) \leq \text{BRev}(D). \tag{82}$$

Consequently, $\Gamma(1, k) = 1$ for every $k \geq 1$.

Proof. Write the one-dimensional type as $v \geq 0$. Let

$$u(v) = \sup_{(q,p) \in M} (vq - p) \quad (83)$$

be the indirect utility. Because $q \in [0, 1]$, $p \geq 0$, and the outside option is present,

$$0 \leq u(v) \leq v, \quad u(0) = 0. \quad (84)$$

As a supremum of affine functions with slopes in $[0, 1]$, u is finite, convex, nondecreasing, and 1-Lipschitz.

We use the following standard one-dimensional convex-function facts in the precise form required. A finite convex function on an interval has a right derivative at every interior point; its right derivative is nondecreasing and right-continuous. A Lipschitz convex function is absolutely continuous on compact intervals, and

$$u(v) - u(0) = \int_0^v u'_+(t) dt. \quad (85)$$

All these hypotheses hold by (84). Define

$$Q(v) = u'_+(v).$$

Then

$$0 \leq Q(v) \leq 1, \quad (86)$$

and Q is nondecreasing and right-continuous.

Let $(q(v), p(v))$ be a selected option at type v . Since this option attains (83), for every $z \geq 0$,

$$u(z) \geq zq(v) - p(v) = u(v) + q(v)(z - v). \quad (87)$$

Thus $q(v)$ is a subgradient of u at v . Taking $z \downarrow v$ in (87) gives

$$q(v) \leq u'_+(v) = Q(v). \quad (88)$$

For $v > 0$,

$$p(v) = vq(v) - u(v) \leq vQ(v) - u(v). \quad (89)$$

At $v = 0$, individual rationality and nonnegative prices give $p(0) = 0$.

The bounded, nondecreasing, right-continuous function Q defines a finite positive Lebesgue–Stieltjes measure μ_Q on $(0, \infty)$, characterized by

$$\mu_Q((a, b]) = Q(b) - Q(a), \quad 0 \leq a < b. \quad (90)$$

Its total mass satisfies

$$\mu_Q((0, \infty)) = \lim_{v \rightarrow \infty} Q(v) - Q(0) \leq 1. \quad (91)$$

The Stieltjes integration-by-parts identity for a bounded monotone right-continuous function and the continuously differentiable function $t \mapsto t$ states that

$$\int_{(0,v]} t dQ(t) = vQ(v) - \int_0^v Q(t) dt. \quad (92)$$

The required boundedness and monotonicity hypotheses are supplied by (86), and (85) turns (92) into

$$\int_{(0,v]} t dQ(t) = vQ(v) - u(v). \quad (93)$$

Combining (89) and (93),

$$p(v) \leq \int_{(0,v]} t dQ(t). \quad (94)$$

Let $V \sim D$. The integrand

$$(t, V) \mapsto t \mathbf{1}_{\{t \leq V\}}$$

is nonnegative and measurable. The probability measure of V and the finite positive measure μ_Q are both σ -finite. Tonelli's theorem for nonnegative measurable functions therefore applies and gives

$$\begin{aligned} \mathbb{E}[p(V)] &\leq \mathbb{E} \left[\int_{(0,V]} t dQ(t) \right] \\ &= \int_{(0,\infty)} t \Pr[V \geq t] dQ(t). \end{aligned} \quad (95)$$

By the definition of bundle revenue in one dimension,

$$t \Pr[V \geq t] \leq \text{BRev}(D) \quad (t > 0).$$

Using the positivity of μ_Q and (91),

$$\mathbb{E}[p(V)] \leq \text{BRev}(D) \int_{(0,\infty)} dQ(t) \leq \text{BRev}(D). \quad (96)$$

Conversely, every posted price t is buy- k incentive compatible and earns

$$t \Pr[V \geq t].$$

Although the supremum defining $\text{BRev}(D)$ need not be attained, taking the supremum over posted prices gives

$$\text{Buy}_k \text{Rev}(D) \geq \text{BRev}(D).$$

Together with (96), this proves

$$\text{Buy}_k \text{Rev}(D) = \text{BRev}(D),$$

and hence $\Gamma(1, k) = 1$. □

8 Conclusion

The buy-two packing lemma and its revenue conversion show that every buy- k mechanism with $k \geq 2$ admits a finite distribution-free bundling approximation. The finite-support construction proves that no such approximation exists for buy-one mechanisms when $n \geq 2$, while the convexity argument proves the exact optimality of bundling for one item. These results establish the complete threshold

$$k_{\min}(1) = 1, \quad k_{\min}(n) = 2 \quad (n \geq 2).$$

The sharper telescoping argument additionally yields the $8n^2$ approximation for $k \geq n$. The optimization of the numerical constants is separate from, and unnecessary for, the resolved finiteness-threshold question.

References

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